BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

# Module 4 Lab Packet.

Respiratory · Lymphatic · Digestive · Abdominal Vessels · Lower Limb · Endocrine · Neck

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|---|---|
| PART A | **Lab structure list**<br>Every structure you can be asked to identify on the Module 4 lab exam, chapter by chapter. |
| PART B | **Charts**<br>Origin, insertion, action, and nerve for every muscle on the list, flow charts for the routes you trace, and reference charts. |
| PART C | **Lab exam questions**<br>The kinds of questions you could be asked at a station, by chapter. |

# How to use this packet

## In lab

Bring Part A to every lab. Check off a structure only when you can find it on the model, the cadaver, or the slide without looking at a label. Say the name out loud to your team as you point to it.

## The charts

For every muscle, the lab exam can ask where it starts (origin), where it ends (insertion), what it does (action), and which nerve supplies it. Cover the chart and quiz yourself one column at a time. Trace each flow chart with your finger on the model, then cover it and say the route from memory.

## The questions

Part C shows the kinds of questions that come with a pinned or pointed structure. Practice them out loud with a partner, with the structure in front of you.

PART A

# Lab structure list.

These are the lab structures and concepts to learn for the lab portion of the Module 4 exam. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

## The Respiratory System

MODELS, CADAVER, AND SLIDES

Competencies 97 to 104

Follow the air. Every structure in this list sits somewhere along one continuous path, and if you can trace the path you can place any structure on it. The thoracic block at the end is the same cadaver dissection the master list asks for, so work the two together.

### ☐ External nose and nasal cavity

STRUCTURES TO IDENTIFY

- ☐ Nares (nostrils)
- ☐ Nasal vestibule
- ☐ Nasal cavity
- ☐ Nasal septum
- ☐ Septal cartilage
- ☐ Perpendicular plate of the ethmoid
- ☐ Vomer
- ☐ Superior nasal concha
- ☐ Middle nasal concha
- ☐ Inferior nasal concha
- ☐ Hard palate
- ☐ Soft palate
- ☐ Choanae

### ☐ Paranasal sinuses

Name the bone each sinus is named for and say where it drains.

STRUCTURES TO IDENTIFY

- ☐ Frontal sinus
- ☐ Ethmoidal air cells
- ☐ Sphenoidal sinus
- ☐ Maxillary sinus

### ☐ Pharynx

Name the epithelium of each region and say whether it carries air, food, or both.

STRUCTURES TO IDENTIFY

- ☐ Nasopharynx
- ☐ Pharyngeal tonsil
- ☐ Opening of the auditory tube
- ☐ Oropharynx
- ☐ Palatine tonsil
- ☐ Lingual tonsil
- ☐ Laryngopharynx
- ☐ Uvula
- ☐ Soft palate

### ☐ Larynx

Identify which single cartilage is elastic rather than hyaline, and which one is the only complete ring.

STRUCTURES TO IDENTIFY

- ☐ Thyroid cartilage
- ☐ Laryngeal prominence
- ☐ Cricoid cartilage
- ☐ Epiglottis
- ☐ Arytenoid cartilages
- ☐ Corniculate cartilages
- ☐ Cuneiform cartilages
- ☐ Cricothyroid ligament
- ☐ Vocal folds (true vocal cords)
- ☐ Vocal ligaments
- ☐ Vestibular folds (false vocal cords)
- ☐ Glottis
- ☐ Hyoid bone

### ☐ Trachea and bronchial tree

Be prepared to say which main bronchus is wider and more vertical, and why that matters.

**STRUCTURES TO IDENTIFY**

- ☐ Trachea
- ☐ Tracheal cartilages (C rings)
- ☐ Trachealis muscle
- ☐ Carina
- ☐ Right main (primary) bronchus
- ☐ Left main (primary) bronchus
- ☐ Lobar (secondary) bronchi
- ☐ Segmental (tertiary) bronchi
- ☐ Bronchioles
- ☐ Terminal bronchioles

### ☐ Lungs and pleurae

**STRUCTURES TO IDENTIFY**

- ☐ Right lung, superior lobe
- ☐ Right lung, middle lobe
- ☐ Right lung, inferior lobe
- ☐ Horizontal fissure
- ☐ Oblique fissure
- ☐ Left lung, superior lobe
- ☐ Left lung, inferior lobe
- ☐ Cardiac notch
- ☐ Lingula
- ☐ Apex of the lung
- ☐ Base of the lung
- ☐ Hilum of the right lung
- ☐ Hilum of the left lung
- ☐ Parietal pleura
- ☐ Visceral pleura
- ☐ Right pleural cavity
- ☐ Left pleural cavity

### ☐ The thorax on the cadaver

Memory trick for the fiber directions: hands in pockets is external, praying hands is internal.

**STRUCTURES TO IDENTIFY**

- ☐ Thoracic cavity
- ☐ Mediastinum
- ☐ Esophagus
- ☐ Diaphragm
- ☐ External intercostals
- ☐ Internal intercostals

Innervation of the diaphragm.

Name the three structures that pass through the diaphragm and the cord level of each: T8

T10

T12

### ☐ Vessels and nerves of the thorax

Give the side and say whether each vessel is an artery or a vein before you name it.

**STRUCTURES TO IDENTIFY**

- ☐ Brachiocephalic trunk
- ☐ Right subclavian artery
- ☐ Left subclavian artery
- ☐ Right common carotid artery
- ☐ Left common carotid artery
- ☐ Right phrenic nerve
- ☐ Left phrenic nerve
- ☐ Right brachiocephalic vein
- ☐ Left brachiocephalic vein
- ☐ Right subclavian vein
- ☐ Left subclavian vein
- ☐ Right internal jugular vein
- ☐ Left internal jugular vein
- ☐ Right pulmonary artery
- ☐ Left pulmonary artery
- ☐ Right pulmonary veins
- ☐ Left pulmonary veins

☐ **The respiratory zone and the alveolus**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

- ☐ Respiratory bronchiole
- ☐ Alveolar duct
- ☐ Alveolar sac
- ☐ Alveolus
- ☐ Pulmonary capillary

☐ **Respiratory slides**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

- ☐ Pseudostratified ciliated columnar epithelium
- ☐ Goblet cells
- ☐ Cilia
- ☐ Hyaline cartilage of the trachea
- ☐ Simple squamous epithelium of the alveolus
- ☐ Lung tissue, low power

☐ **Also identify**

STRUCTURES TO IDENTIFY

- ☐ Larynx
- ☐ Hilum
- ☐ Pleural cavities

Trace a breath of air from the nostril to an alveolus, naming every structure it passes.

CHAPTER 2

# The Lymphatic System

MODELS, CADAVER, AND SLIDES

Competencies 105 to 108

A one-way drainage network. The master list names the spleen and the three tonsils outright; the vessels, the node, and the organ list come from the lymphatic chapter of this module. Learn the route first and the parts fall onto it.

☐ **Lymphatic vessels, from the tissue spaces to the veins**

Say which duct drains the upper right quadrant and which drains the rest of the body.

STRUCTURES TO IDENTIFY

- ☐ Lymphatic capillaries
- ☐ Lacteals
- ☐ Lymphatic collecting vessels
- ☐ Lymphatic valves
- ☐ Lymphatic trunks
- ☐ Right lymphatic duct
- ☐ Thoracic duct
- ☐ Cisterna chyli
- ☐ Subclavian veins
- ☐ Internal jugular veins

☐ **The lymph node**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

- ☐ Capsule
- ☐ Cortex
- ☐ Medulla
- ☐ Afferent lymphatic vessels
- ☐ Efferent lymphatic vessels
- ☐ Hilum
- ☐ Cervical nodes
- ☐ Axillary nodes
- ☐ Inguinal nodes

☐ **The lymphatic organs**

Sort each one as a primary or a secondary lymphatic organ.

STRUCTURES TO IDENTIFY

- ☐ Thymus
- ☐ Lymph nodes
- ☐ Spleen

☐ **The spleen**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

- ☐ Spleen

☐ **The tonsils**

Name the region of the pharynx each tonsil guards.

STRUCTURES TO IDENTIFY

- ☐ Pharyngeal tonsil
- ☐ Palatine tonsil
- ☐ Lingual tonsil

Which two lymphatic organs are primary, and what happens in each?

Trace a drop of interstitial fluid from a tissue space back to the bloodstream.

CHAPTER 3

# The Alimentary Canal

THE CADAVER, THE TORSO MODEL, AND SLIDES

Competencies 109 to 119

One continuous tube from the mouth to the anus. Work the peritoneum first, because the folds are what you have to move aside to reach the organs, and then take the canal region by region.

☐ **Body cavities, serous membranes, and the omenta**

STRUCTURES TO IDENTIFY

- ☐ Abdominal cavity
- ☐ Pelvic cavity
- ☐ Peritoneal cavity
- ☐ Parietal peritoneum
- ☐ Visceral peritoneum
- ☐ Retroperitoneum
- ☐ Greater omentum
- ☐ Lesser omentum
- ☐ Falciform ligament
- ☐ Mesentery
- ☐ Mesocolon

Name the retroperitoneal structures. Memory device: SADPUCKER.

☐ **The wall of the canal, in section**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

- ☐ Mucosa
- ☐ Epithelium
- ☐ Lamina propria
- ☐ Muscularis mucosae
- ☐ Submucosa
- ☐ Muscularis, inner circular layer
- ☐ Muscularis, outer longitudinal layer
- ☐ Serosa
- ☐ Adventitia
- ☐ Lumen

☐ **Mouth, pharynx, and esophagus**

Say which end of the esophagus carries an adventitia rather than a serosa.

STRUCTURES TO IDENTIFY

- ☐ Oral cavity
- ☐ Hard palate
- ☐ Soft palate
- ☐ Uvula
- ☐ Palatine tonsil
- ☐ Lingual tonsil
- ☐ Nasopharynx
- ☐ Oropharynx
- ☐ Laryngopharynx
- ☐ Esophagus
- ☐ Esophageal hiatus
- ☐ Upper esophageal sphincter
- ☐ Lower (cardiac) esophageal sphincter

☐ **The stomach**

The stomach carries one more muscle layer than the rest of the tube. Name it and give its position.

STRUCTURES TO IDENTIFY

- ☐ Cardiac region
- ☐ Fundus
- ☐ Body
- ☐ Pyloric region
- ☐ Pyloric antrum
- ☐ Pyloric canal
- ☐ Pylorus
- ☐ Pyloric sphincter
- ☐ Cardiac sphincter
- ☐ Lesser curvature
- ☐ Greater curvature
- ☐ Gastric rugae

☐ **The small intestine**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

- ☐ Duodenum
- ☐ Major duodenal papilla
- ☐ Jejunum
- ☐ Ileum
- ☐ Ileocecal valve
- ☐ Ileocecal sphincter
- ☐ Mesentery
- ☐ Circular folds (plicae circulares)
- ☐ Villi
- ☐ Lacteal
- ☐ Microvilli (brush border)
- ☐ Sphincter of Oddi (hepatopancreatic ampulla)

☐ **The large intestine**

Name the three features that belong to the large intestine and to no other part of the canal.

STRUCTURES TO IDENTIFY

- ☐ Cecum
- ☐ Vermiform appendix
- ☐ Ascending colon
- ☐ Hepatic flexure (right colic flexure)
- ☐ Transverse colon
- ☐ Splenic flexure (left colic flexure)
- ☐ Descending colon
- ☐ Sigmoid colon
- ☐ Rectum
- ☐ Anal canal
- ☐ Internal anal sphincter
- ☐ External anal sphincter
- ☐ Teniae coli
- ☐ Haustra
- ☐ Omental appendices

☐ **Also identify**

STRUCTURES TO IDENTIFY

☐ Stomach ☐ Small intestine ☐ Large intestine

Trace a bite of food from the mouth to the anus, naming every region it passes through.

CHAPTER 4

## The Accessory Digestive Organs

THE CADAVER, THE TORSO MODEL, AND SLIDES

Competencies 120 to 125

The organs that sit alongside the tube and empty into it. Two questions on this sheet are hers outright: the fetal remnant behind the ligamentum teres, and the path of bile.

☐ **The tongue and the salivary glands**

Give the location of each of the three large salivary glands and say where its duct opens.

STRUCTURES TO IDENTIFY

☐ Tongue ☐ Papillae ☐ Submandibular gland
☐ Lingual frenulum ☐ Parotid gland ☐ Sublingual gland

☐ **The liver**

STRUCTURES TO IDENTIFY

☐ Right lobe ☐ Caudate lobe ☐ Lesser omentum
☐ Left lobe ☐ Falciform ligament
☐ Quadrate lobe ☐ Ligamentum teres (round ligament)

Ligamentum teres. It is the remnant of which fetal structure?

☐ **Liver tissue, on the slide and the model**

**Function** _____ **Location** _____

**Special structures** _____

STRUCTURES TO IDENTIFY

☐ Hepatocytes ☐ Portal triad ☐ Branch of the hepatic portal vein
☐ Hepatic sinusoids ☐ Bile duct ☐ Central vein
☐ Bile canaliculi ☐ Branch of the hepatic artery

☐ **The gallbladder and the biliary tree**

STRUCTURES TO IDENTIFY

☐ Gallbladder ☐ Right hepatic duct ☐ Common bile duct
☐ Fundus of the gallbladder ☐ Left hepatic duct ☐ Hepatopancreatic ampulla
☐ Body of the gallbladder ☐ Common hepatic duct ☐ Major duodenal papilla
☐ Neck of the gallbladder ☐ Cystic duct ☐ Sphincter of Oddi

☐ **The pancreas**

Say which part of the pancreas is exocrine and which is endocrine, and which duct joins the common bile duct.

STRUCTURES TO IDENTIFY

- ☐ Head of the pancreas
- ☐ Body of the pancreas
- ☐ Tail of the pancreas
- ☐ Main pancreatic duct
- ☐ Accessory pancreatic duct
- ☐ Hepatopancreatic ampulla
- ☐ Major duodenal papilla
- ☐ Acini

☐ **Also identify**

STRUCTURES TO IDENTIFY

- ☐ Pancreas
- ☐ Liver

Trace a drop of bile from the hepatocyte that made it to the duodenum.

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CHAPTER 5

## Vessels of the Abdomen and Pelvis

CADAVER AND TORSO MODEL

Competency 126

The master list groups these as arteries then veins, with the hepatic portal system kept separate. Learn the three unpaired branches of the abdominal aorta in order from the top, because their order is what the tag is testing.

☐ **Arteries of the abdomen**

Name the three unpaired visceral branches of the abdominal aorta in order from superior to inferior, and say what each one supplies.

STRUCTURES TO IDENTIFY

- ☐ Abdominal aorta
- ☐ Celiac trunk
- ☐ Left gastric artery
- ☐ Splenic artery
- ☐ Common hepatic artery
- ☐ Superior mesenteric artery
- ☐ Renal arteries
- ☐ Gonadal (testicular or ovarian) arteries
- ☐ Inferior mesenteric artery
- ☐ Right common iliac artery
- ☐ Left common iliac artery

☐ **Arteries of the pelvis**

STRUCTURES TO IDENTIFY

- ☐ Internal iliac artery
- ☐ External iliac artery
- ☐ Medial umbilical ligament

☐ **Veins of the abdomen and pelvis**

STRUCTURES TO IDENTIFY

- ☐ Inferior vena cava
- ☐ Renal veins
- ☐ Gonadal (testicular or ovarian) veins
- ☐ Right common iliac vein
- ☐ Left common iliac vein
- ☐ Internal iliac vein
- ☐ External iliac vein

☐ **The hepatic portal system**

Say what makes this a portal system, and name the two capillary beds the blood passes through.

STRUCTURES TO IDENTIFY

- ☐ Hepatic portal vein
- ☐ Splenic vein
- ☐ Superior mesenteric vein
- ☐ Inferior mesenteric vein
- ☐ Hepatic veins

Trace a molecule of glucose absorbed by the small intestine from the intestinal capillary to the right atrium.

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CHAPTER 6

## Muscles, Nerves and Vessels of the Lower Limb

MODELS AND CADAVER

Competency 127

Muscles first, then the nerves and vessels that run with them. For every muscle, be able to name it, find it on the models and the cadaver, and give its origin, insertion, and action. Use right and left, and artery and vein, when you name a vessel.

☐ **Anterior, medial and lateral thigh**

Give the origin, insertion, and action of each muscle.

STRUCTURES TO IDENTIFY

- ☐ Iliopsoas (iliacus and psoas major)
- ☐ Rectus femoris
- ☐ Vastus lateralis
- ☐ Vastus intermedius
- ☐ Vastus medialis
- ☐ Sartorius
- ☐ Gracilis
- ☐ Adductor magnus
- ☐ Adductor longus
- ☐ Adductor brevis
- ☐ Pectineus
- ☐ Tensor fasciae latae
- ☐ Iliotibial (IT) band
- ☐ Iliacus
- ☐ Psoas major
- ☐ Quadriceps femoris

☐ **Gluteal region and posterior thigh**

Give the origin, insertion, and action of each muscle.

STRUCTURES TO IDENTIFY

- ☐ Gluteus maximus
- ☐ Gluteus medius
- ☐ Biceps femoris
- ☐ Semitendinosus
- ☐ Semimembranosus
- ☐ Hamstrings

☐ **The leg**

Give the origin, insertion, and action of each muscle.

STRUCTURES TO IDENTIFY

- ☐ Tibialis anterior
- ☐ Extensor hallucis longus
- ☐ Extensor digitorum longus
- ☐ Fibularis brevis
- ☐ Fibularis longus
- ☐ Tibialis posterior
- ☐ Flexor digitorum longus
- ☐ Flexor hallucis longus
- ☐ Gastrocnemius
- ☐ Soleus
- ☐ Plantaris

☐ **Nerves and vessels of the lower limb**

STRUCTURES TO IDENTIFY

- ☐ Femoral nerve
- ☐ Sciatic nerve
- ☐ Tibial nerve
- ☐ Common fibular nerve
- ☐ Femoral artery
- ☐ Femoral vein
- ☐ Great saphenous vein
- ☐ Posterior tibial artery

CHAPTER 7

## The Endocrine System

TORSO MODEL, CADAVER, AND SLIDES

Competencies 128 to 132

The endocrine glands are scattered, so this sheet is a tour of locations first. The master list names the thyroid gland in the neck and the endocrine pancreas and the suprarenal glands in the abdomen; the rest come from the endocrine chapter of this module.

### ☐ Locating the glands

Give the location of each gland before you name it.

**STRUCTURES TO IDENTIFY**

- ☐ Hypothalamus
- ☐ Pituitary gland
- ☐ Pineal gland
- ☐ Thyroid gland
- ☐ Parathyroid glands
- ☐ Thymus
- ☐ Adrenal (suprarenal) glands
- ☐ Pancreas
- ☐ Ovaries
- ☐ Testes

### ☐ The thyroid and parathyroid glands

**STRUCTURES TO IDENTIFY**

- ☐ Thyroid gland, right lobe
- ☐ Thyroid gland, left lobe
- ☐ Isthmus
- ☐ Parathyroid glands
- ☐ Trachea
- ☐ Larynx

### ☐ Thyroid tissue, on the slide

**Function** _____ **Location** _____

**Special structures** _____

**STRUCTURES TO IDENTIFY**

- ☐ Thyroid follicles
- ☐ Colloid
- ☐ Follicular cells
- ☐ Parafollicular (C) cells

### ☐ The adrenal gland

**Function** _____ **Location** _____

**Special structures** _____

**STRUCTURES TO IDENTIFY**

- ☐ Adrenal (suprarenal) gland
- ☐ Adrenal cortex
- ☐ Zona glomerulosa
- ☐ Zona fasciculata
- ☐ Zona reticularis
- ☐ Adrenal medulla
- ☐ Kidney

### ☐ The endocrine pancreas

Say which of these two clusters is endocrine and which is exocrine.

**STRUCTURES TO IDENTIFY**

- ☐ Pancreas
- ☐ Pancreatic islets
- ☐ Acini

Which gland sits on the superior pole of each kidney, and what are its two parts made of?

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CHAPTER 8

## The Anterior Neck

**MODELS AND CADAVER**

These are on the Unit 4 worksheets of the master list.

☐ **Anterior neck**

Platysma and sternocleidomastoid are on Exam 3 this time.

**STRUCTURES TO IDENTIFY**

- ☐ Sternocleidomastoid
- ☐ Common carotid artery
- ☐ Vertebral artery
- ☐ Internal jugular vein

PART B

# Charts.

Muscle charts first, grouped the way you will find the muscles in lab, then the flow charts and reference charts.

SCAN TO DRILL THESE CHARTS

**Interactive Muscle Charts: I, O, A**

Quiz yourself on the origin, insertion, action, and nerve for each muscle on your phone.

medmasterscollaborative.com/muscle-charts-i-o-a-inn

## Muscles of the anterior, medial and lateral thigh

Anterior compartment: femoral nerve. Medial compartment: obturator nerve. The iliotibial (IT) band is not a muscle; it is a thick band of fascia from the iliac crest to the lateral condyle of the tibia, pulled on by the tensor fasciae latae and gluteus maximus.

| MUSCLE | ORIGIN | INSERTION | ACTION | NERVE |
|---|---|---|---|---|
| **Iliopsoas (iliacus and psoas major)** | Iliac fossa (iliacus) and vertebrae T12 to L5 (psoas major) | Lesser trochanter of the femur | Flexes the thigh at the hip | Femoral nerve (iliacus) and ventral rami of L1 to L3 (psoas major) |
| **Iliacus** | Iliac fossa | Lesser trochanter of the femur, with the psoas major | Flexes the thigh at the hip | Femoral nerve |
| **Psoas major** | Bodies and transverse processes of vertebrae T12 to L5 | Lesser trochanter of the femur, with the iliacus | Flexes the thigh at the hip; flexes the trunk when the thigh is fixed | Ventral rami of L1 to L3 |
| **Quadriceps femoris (rectus femoris and the three vasti)** | Anterior inferior iliac spine (rectus femoris) and the shaft of the femur (vasti) | Tibial tuberosity, through the quadriceps tendon, patella and patellar ligament | Extends the knee; the rectus femoris also flexes the thigh | Femoral nerve |
| **Rectus femoris** | Anterior inferior iliac spine | Tibial tuberosity, through the quadriceps tendon, patella and patellar ligament | Extends the knee and flexes the thigh at the hip | Femoral nerve |
| **Vastus lateralis** | Greater trochanter and the lateral lip of the linea aspera | Tibial tuberosity, through the quadriceps tendon, patella and patellar ligament | Extends the knee | Femoral nerve |
| **Vastus intermedius** | Anterior and lateral surfaces of the femoral shaft | Tibial tuberosity, through the quadriceps tendon, patella and patellar ligament | Extends the knee | Femoral nerve |
| **Vastus medialis** | Intertrochanteric line and the medial lip of the linea aspera | Tibial tuberosity, through the quadriceps tendon, patella and patellar ligament | Extends the knee | Femoral nerve |

| MUSCLE | ORIGIN | INSERTION | ACTION | NERVE |
|---|---|---|---|---|
| **Sartorius** | Anterior superior iliac spine | Medial surface of the upper tibia (pes anserinus) | Flexes, abducts and laterally rotates the thigh; flexes the knee | Femoral nerve |
| **Gracilis** | Body and inferior ramus of the pubis | Medial surface of the upper tibia (pes anserinus) | Adducts the thigh; flexes the knee and medially rotates the leg | Obturator nerve |
| **Adductor magnus** | Inferior ramus of the pubis, ramus of the ischium and the ischial tuberosity | Linea aspera and the adductor tubercle of the femur | Adducts the thigh; the upper part also flexes and medially rotates it, and the part from the ischial tuberosity extends it | Obturator nerve, and the tibial part of the sciatic nerve for the part from the ischial tuberosity |
| **Adductor longus** | Body of the pubis, near the pubic symphysis | Middle of the linea aspera | Adducts, flexes and medially rotates the thigh | Obturator nerve |
| **Adductor brevis** | Body and inferior ramus of the pubis | Upper part of the linea aspera | Adducts, flexes and medially rotates the thigh | Obturator nerve |
| **Pectineus** | Superior ramus of the pubis (pectineal line) | Pectineal line of the femur, just below the lesser trochanter | Adducts and flexes the thigh | Femoral nerve (sometimes also the obturator nerve); it sits in the medial compartment even though the femoral nerve supplies it |
| **Tensor fasciae latae** | Anterior iliac crest and the anterior superior iliac spine | Iliotibial (IT) band, which runs to the lateral condyle of the tibia | Flexes, abducts and medially rotates the thigh; tightens the IT band to steady the knee | Superior gluteal nerve |

## Muscles of the gluteal region and posterior thigh

The hamstrings are the biceps femoris, semitendinosus and semimembranosus. Each crosses the back of the hip and the knee, so each extends the thigh and flexes the knee (the short head of the biceps femoris crosses only the knee).

| MUSCLE | ORIGIN | INSERTION | ACTION | NERVE |
|---|---|---|---|---|
| **Gluteus maximus** | Posterior surface of the ilium, the sacrum and the coccyx | Gluteal tuberosity of the femur and the iliotibial (IT) band | Extends the thigh at the hip, the main extensor for climbing and rising from a chair; laterally rotates the thigh | Inferior gluteal nerve |
| **Gluteus medius** | Outer (lateral) surface of the ilium | Greater trochanter of the femur | Abducts and medially rotates the thigh; keeps the pelvis level when the opposite foot is off the ground | Superior gluteal nerve |
| **Hamstrings (biceps femoris, semitendinosus, semimembranosus)** | Ischial tuberosity (and the linea aspera for the short head of the biceps femoris) | Head of the fibula (biceps femoris) and the medial upper tibia (semitendinosus and semimembranosus) | Extend the thigh at the hip and flex the knee | Sciatic nerve |
| **Biceps femoris** | Long head: ischial tuberosity. Short head: linea aspera of the femur | Head of the fibula (and the lateral condyle of the tibia) | Flexes the knee and laterally rotates the leg; the long head also extends the thigh | Sciatic nerve (long head: tibial part; short head: common fibular part) |
| **Semitendinosus** | Ischial tuberosity | Medial surface of the upper tibia (pes anserinus) | Flexes the knee and extends the thigh; medially rotates the leg | Sciatic nerve (tibial part) |
| **Semimembranosus** | Ischial tuberosity | Posterior surface of the medial condyle of the tibia | Flexes the knee and extends the thigh; medially rotates the leg | Sciatic nerve (tibial part) |

## Muscles of the leg

Anterior compartment: deep fibular nerve. Lateral compartment: superficial fibular nerve. Posterior compartment: tibial nerve. Peroneus means the same as fibularis.

| MUSCLE | ORIGIN | INSERTION | ACTION | NERVE |
|---|---|---|---|---|
| **Tibialis anterior** | Lateral condyle and upper lateral shaft of the tibia, and the interosseous membrane | Medial cuneiform and the base of metatarsal 1 | Dorsiflexes and inverts the foot | Deep fibular nerve |
| **Extensor hallucis longus** | Middle of the anterior fibula and the interosseous membrane | Distal phalanx of the great toe | Extends the great toe; dorsiflexes the foot | Deep fibular nerve |
| **Extensor digitorum longus** | Lateral condyle of the tibia, upper anterior fibula and the interosseous membrane | Middle and distal phalanges of toes 2 to 5 | Extends toes 2 to 5; dorsiflexes the foot | Deep fibular nerve |
| **Fibularis longus** | Head and upper lateral shaft of the fibula | Medial cuneiform and the base of metatarsal 1, by a tendon that crosses the sole | Everts and plantar flexes the foot; helps support the arches | Superficial fibular nerve |
| **Fibularis brevis** | Lower lateral shaft of the fibula | Base of metatarsal 5 | Everts and plantar flexes the foot | Superficial fibular nerve |
| **Gastrocnemius** | Medial and lateral condyles of the femur | Calcaneus, by the calcaneal (Achilles) tendon | Plantar flexes the foot and flexes the knee | Tibial nerve |
| **Soleus** | Head and upper posterior shaft of the fibula and the soleal line of the tibia | Calcaneus, by the calcaneal (Achilles) tendon | Plantar flexes the foot | Tibial nerve |
| **Plantaris** | Femur, just above the lateral condyle | Calcaneus, beside the calcaneal tendon | Weakly helps plantar flex the foot and flex the knee | Tibial nerve |
| **Tibialis posterior** | Interosseous membrane and the posterior surfaces of the tibia and fibula | Navicular, plus the cuneiforms, cuboid and bases of metatarsals 2 to 4 | Inverts and plantar flexes the foot | Tibial nerve |
| **Flexor digitorum longus** | Posterior surface of the tibia | Distal phalanges of toes 2 to 5 | Flexes toes 2 to 5; helps plantar flex and invert the foot | Tibial nerve |
| **Flexor hallucis longus** | Lower posterior fibula and the interosseous membrane | Distal phalanx of the great toe | Flexes the great toe; helps plantar flex and invert the foot | Tibial nerve |

## Muscle of the anterior neck

The sternocleidomastoid divides the neck into its anterior and posterior triangles.

| MUSCLE | ORIGIN | INSERTION | ACTION | NERVE |
|---|---|---|---|---|
| **Sternocleidomastoid** | Manubrium of the sternum and the medial clavicle | Mastoid process of the temporal bone | Both sides: flex the neck. One side: tilts the head to the same side and turns the face to the opposite side | Accessory nerve (CN XI) |

## The path of air from the nostrils to the alveoli

The conducting zone carries air and ends at the terminal bronchioles. The respiratory zone, where gases are exchanged, starts at the respiratory bronchioles.

1. **Nares (nostrils)** the openings where air enters the nose
2. **Nasal vestibule** just inside the nostrils; skin and coarse hairs filter large particles
3. **Nasal cavity** air swirls past the superior, middle and inferior nasal conchae and is warmed, moistened and filtered
4. **Choanae** the posterior openings of the nasal cavity into the pharynx
5. **Nasopharynx** behind the nasal cavity, above the soft palate; carries air only
6. **Oropharynx** behind the oral cavity; carries air and food
7. **Laryngopharynx** from the epiglottis down to where the air and food paths divide
8. **Larynx** air passes the epiglottis and goes through the glottis, the opening between the vocal folds
9. **Trachea** held open by C-shaped rings of hyaline cartilage; forks at the carina
10. **Right or left main (primary) bronchus** one to each lung; the right is wider and more vertical
11. **Lobar (secondary) bronchi** one to each lobe: three on the right, two on the left
12. **Segmental (tertiary) bronchi** one to each bronchopulmonary segment
13. **Bronchioles** smooth muscle walls and no cartilage
14. **Terminal bronchioles** the last airways of the conducting zone
15. **Respiratory bronchioles** the first airways of the respiratory zone, with a few alveoli budding from the wall
16. **Alveolar ducts** passages lined with alveoli
17. **Alveolar sacs** clusters of alveoli sharing a common space
18. **Alveoli** gas exchange with the pulmonary capillaries across the simple squamous wall

## Lymph, from tissue fluid back to the blood

Lymph flows one way, toward the heart. Valves, skeletal muscle and breathing keep it moving.

1. **Interstitial fluid** fluid that leaks out of the blood capillaries into the spaces between cells
2. **Lymphatic capillaries** blind-ended tubes that take the fluid in; now it is called lymph. In the villi of the small intestine they are lacteals, which take in fat and make milky chyle
3. **Lymphatic collecting vessels** carry lymph onward; many lymphatic valves stop it flowing backward
4. **Lymph nodes** lymph enters through afferent lymphatic vessels, is filtered, and leaves through efferent lymphatic vessels at the hilum
5. **Lymphatic trunks** collecting vessels merge into large trunks that each drain a region
6. **Right lymphatic duct or thoracic duct** the right lymphatic duct drains the right arm and the right side of the head, neck and thorax. The thoracic duct begins at the cisterna chyli and drains the rest of the body
7. **Subclavian vein** each duct empties where the subclavian vein meets the internal jugular vein; the right duct on the right side, the thoracic duct on the left

## The path of food from the mouth to the anus

1. **Oral cavity** food is chewed and mixed with saliva
2. **Oropharynx** the swallowed food passes behind the mouth
3. **Laryngopharynx** the epiglottis covers the larynx so food goes to the esophagus
4. **Upper esophageal sphincter** a ring of skeletal muscle at the top of the esophagus

5. **Esophagus** carries food down through the esophageal hiatus of the diaphragm
6. **Lower (cardiac) esophageal sphincter** guards the entrance to the stomach
7. **Stomach** food enters at the cardiac region, is churned in the body, and moves through the pyloric antrum, pyloric canal and pylorus
8. **Pyloric sphincter** lets small amounts of food into the small intestine
9. **Duodenum** the first and shortest part; bile and pancreatic juice enter at the major duodenal papilla
10. **Jejunum** the middle part of the small intestine
11. **Ileum** the longest part of the small intestine
12. **Ileocecal valve** where the ileum joins the large intestine
13. **Cecum** the pouch at the start of the large intestine; the vermiform appendix hangs from it
14. **Ascending colon** up the right side
15. **Hepatic flexure (right colic flexure)** the bend under the liver
16. **Transverse colon** across the abdomen
17. **Splenic flexure (left colic flexure)** the bend near the spleen
18. **Descending colon** down the left side
19. **Sigmoid colon** the S-shaped part in the pelvis
20. **Rectum** the straight part in front of the sacrum
21. **Anal canal** guarded by the internal and external anal sphincters
22. **Anus** where feces leave the body

## Blood flow through the liver

Two supplies enter the liver: the hepatic portal vein brings nutrient-rich blood from the digestive organs and the hepatic artery brings oxygen-rich blood. They mix in the sinusoids.

1. **Inferior mesenteric vein** drains the descending colon, sigmoid colon and rectum; usually empties into the splenic vein
2. **Splenic vein** drains the spleen and runs behind the pancreas
3. **Superior mesenteric vein** drains the small intestine, cecum, ascending colon and most of the transverse colon; joins the splenic vein behind the pancreas
4. **Hepatic portal vein** formed where the superior mesenteric and splenic veins join; carries nutrient-rich blood into the liver. It is a portal system because the blood passes through two capillary beds in a row: first the capillaries of the digestive organs, then the hepatic sinusoids
5. **Hepatic artery** the common hepatic artery from the celiac trunk continues to the liver as the hepatic artery proper; it brings oxygen-rich blood
6. **Portal triad** a branch of the hepatic portal vein and a branch of the hepatic artery enter at the corner of each lobule, beside a bile duct
7. **Hepatic sinusoids** the two bloods mix and flow past the plates of hepatocytes toward the center of the lobule
8. **Central vein** collects the blood in the center of each lobule
9. **Hepatic veins** carry the blood out of the liver
10. **Inferior vena cava** the hepatic veins empty into it, and it carries the blood to the right atrium

## Bile flow from the liver to the duodenum

Bile flows toward the portal triad, the opposite way from the blood, which flows toward the central vein.

1. **Hepatocytes** liver cells make bile
2. **Bile canaliculi** tiny channels between the hepatocytes collect it
3. **Bile duct** in the portal triad at the corner of each lobule
4. **Right and left hepatic ducts** drain the right and left sides of the liver
5. **Common hepatic duct** formed where the right and left hepatic ducts join
6. **Cystic duct** between meals, bile backs up the cystic duct into the gallbladder
7. **Gallbladder** stores and concentrates bile, then contracts after a fatty meal and sends it back out the cystic duct
8. **Common bile duct** formed where the cystic duct joins the common hepatic duct
9. **Hepatopancreatic ampulla** the common bile duct joins the main pancreatic duct here
10. **Sphincter of Oddi** the muscle ring that opens to let bile and pancreatic juice through
11. **Major duodenal papilla** the opening on the duodenal wall
12. **Duodenum** bile breaks fat into small droplets

## The branches of the abdominal aorta, top to bottom

The three unpaired branches (celiac trunk, superior mesenteric and inferior mesenteric arteries) supply the digestive organs. The paired branches charted here supply the kidneys and gonads.

1. **Abdominal aorta** begins where the aorta passes through the aortic hiatus of the diaphragm
2. **Celiac trunk** unpaired; splits into the left gastric, splenic and common hepatic arteries to supply the stomach, spleen, liver, gallbladder and parts of the pancreas and duodenum
3. **Superior mesenteric artery** unpaired; supplies the small intestine, cecum, appendix, ascending colon and most of the transverse colon
4. **Renal arteries** paired; arise just below the superior mesenteric artery and supply the kidneys
5. **Gonadal (testicular or ovarian) arteries** paired; supply the testes or the ovaries
6. **Inferior mesenteric artery** unpaired; supplies the end of the transverse colon, the descending colon, sigmoid colon and upper rectum
7. **Right and left common iliac arteries** the aorta ends by splitting into these two
8. **Internal iliac artery** a branch of each common iliac artery; supplies the pelvic organs, such as the bladder and the uterus or prostate
9. **External iliac artery** the other branch of each common iliac artery; becomes the femoral artery as it passes under the inguinal ligament into the thigh

## The regions of the pharynx

The palatine tonsils sit in the oropharynx, on the side walls between the palatal arches.

| REGION | LOCATION | CARRIES | EPITHELIUM | STRUCTURES TO IDENTIFY HERE |
|---|---|---|---|---|
| **Nasopharynx** | Behind the nasal cavity, above the soft palate | Air only | Pseudostratified ciliated columnar | Pharyngeal tonsil, opening of the auditory tube |
| **Oropharynx** | Behind the oral cavity, from the soft palate to the epiglottis | Air and food | Stratified squamous | Palatine tonsil, lingual tonsil |
| **Laryngopharynx** | From the epiglottis down to the esophagus | Air and food | Stratified squamous | The point where air goes to the larynx and food goes to the esophagus |

## The cartilages of the larynx

Three cartilages are single and three are paired, nine in all.

| CARTILAGE | SINGLE OR PAIRED | TYPE OF CARTILAGE | WHAT TO KNOW |
|---|---|---|---|
| **Thyroid cartilage** | Single | Hyaline | The largest; its laryngeal prominence is the Adam's apple |
| **Cricoid cartilage** | Single | Hyaline | Ring-shaped, just below the thyroid cartilage; the only complete ring in the airway. The cricothyroid ligament joins the two |
| **Epiglottis** | Single | Elastic | Spoon-shaped flap that folds over the laryngeal inlet during swallowing |
| **Arytenoid cartilages** | Paired | Hyaline | Anchor and move the vocal folds |
| **Corniculate cartilages** | Paired | Elastic | Small cartilages sitting on top of the arytenoids |
| **Cuneiform cartilages** | Paired | Elastic | Small cartilages in the fold between the arytenoid and the epiglottis |

## The two lungs compared

| LUNG | LOBES | FISSURES | OTHER FEATURES |
|---|---|---|---|
| **Right lung** | Superior, middle and inferior | Horizontal fissure and oblique fissure | Larger; its main bronchus is wider and more vertical |
| **Left lung** | Superior and inferior | Oblique fissure only | Smaller; cardiac notch for the heart, with the lingula below it on the superior lobe |

## The diaphragm and its openings

These are vertebral levels. The diaphragm is supplied by the phrenic nerve (C3, C4, C5).

| OPENING | VERTEBRAL LEVEL | WHAT PASSES THROUGH |
|---|---|---|
| **Caval opening** | T8 | Inferior vena cava |
| **Esophageal hiatus** | T10 | Esophagus (with the vagus nerves) |
| **Aortic hiatus** | T12 | Aorta (with the thoracic duct) |

## The lymphatic organs

Primary organs are where lymphocytes mature. Secondary organs are where mature lymphocytes meet invaders.

| ORGAN | PRIMARY OR SECONDARY | LOCATION | WHAT IT DOES |
|---|---|---|---|
| **Thymus** | Primary | Mediastinum, above the heart and behind the sternum | T cells mature here; large in children, shrinks after puberty |
| **Red bone marrow** | Primary | Spongy bone | All lymphocytes form here, and B cells mature here |

| ORGAN | PRIMARY OR SECONDARY | LOCATION | WHAT IT DOES |
|---|---|---|---|
| **Lymph nodes** | Secondary | Along the collecting vessels; clusters in the cervical, axillary and inguinal regions | Filter lymph |
| **Spleen** | Secondary | Upper left abdomen against the diaphragm, behind the stomach | Filters blood and removes old red blood cells |
| **Pharyngeal tonsil** | Secondary | Posterior wall of the nasopharynx | Guards inhaled air |
| **Palatine tonsil** | Secondary | Side walls of the oropharynx | Guards the throat; the ones usually removed in a tonsillectomy |
| **Lingual tonsil** | Secondary | Base of the tongue, in the oropharynx | Guards swallowed material |

## The lymph node

Lymph enters on the convex side and leaves at the hilum.

| PART | WHAT IT IS |
|---|---|
| **Capsule** | Dense connective tissue covering the node |
| **Cortex** | Outer region with follicles of lymphocytes |
| **Medulla** | Inner region of cords and sinuses lined with macrophages |
| **Afferent lymphatic vessels** | Several vessels that bring lymph into the node through the capsule |
| **Efferent lymphatic vessels** | Fewer vessels that carry filtered lymph out at the hilum |
| **Hilum** | The indented side where the artery enters and the vein and efferent vessels leave |

## The wall of the alimentary canal, deep to superficial

The stomach adds an inner oblique layer of muscle, so it has three muscle layers.

| LAYER | PARTS TO IDENTIFY | WHAT IT IS |
|---|---|---|
| **Mucosa** | Epithelium, lamina propria, muscularis mucosae | The inner lining that touches the food in the lumen |
| **Submucosa** | | Connective tissue with blood vessels, lymphatic vessels and the submucosal nerve plexus |
| **Muscularis** | Inner circular layer, outer longitudinal layer | Mostly smooth muscle (skeletal in the upper esophagus) that mixes and moves food, with the myenteric nerve plexus between the layers |
| **Serosa or adventitia** | | Serosa is the visceral peritoneum. The esophagus above the diaphragm has a fibrous adventitia instead |

## The peritoneum and its folds

**Parietal peritoneum**
The layer that lines the wall of the abdominal cavity

**Visceral peritoneum**
The layer that covers the organs; it is their serosa

**Peritoneal cavity**
The thin, fluid-filled space between the parietal and visceral layers

**Retroperitoneal**
An organ behind the peritoneum, against the posterior abdominal wall

**Greater omentum**
A fatty apron that hangs from the greater curvature of the stomach over the intestines and attaches to the transverse colon

**Falciform ligament**
Attaches the liver to the anterior abdominal wall and the diaphragm

**Mesocolon**
Binds the transverse and sigmoid colon to the posterior abdominal wall

**Intraperitoneal**
An organ wrapped in visceral peritoneum and hanging in the peritoneal cavity, such as the stomach, jejunum and ileum

**SADPUCKER**
The retroperitoneal organs: Suprarenal (adrenal) glands, Aorta and inferior vena cava, Duodenum (most of it), Pancreas (except the tail), Ureters, Colon (ascending and descending), Kidneys, Esophagus, Rectum

**Lesser omentum**
Connects the lesser curvature of the stomach and the duodenum to the liver

**Mesentery**
A fan-shaped fold that binds the jejunum and ileum to the posterior abdominal wall

## Features of the stomach and intestines

**Gastric rugae**
Folds of the stomach lining that flatten as the stomach fills

**Circular folds (plicae circulares)**
Permanent ridges of the small intestine wall that increase surface area

**Microvilli (brush border)**
Tiny projections on each absorptive cell

**Haustra**
The pouches of the colon, formed by the pull of the teniae coli

**Internal anal sphincter**
Smooth muscle; involuntary

**Lesser and greater curvature**
The short, concave medial (right) border and the long, convex lateral (left) border of the stomach

**Villi**
Finger-like projections of the mucosa; each holds a lacteal

**Teniae coli**
Three bands of longitudinal muscle running along the colon

**Omental appendices**
Small fat-filled pouches of peritoneum on the teniae coli

**External anal sphincter**
Skeletal muscle; voluntary

## The salivary glands

| GLAND | LOCATION | WHERE ITS DUCT OPENS |
|---|---|---|
| **Parotid gland** | In front of and below the ear, over the masseter | Inside the cheek, beside the second upper molar |
| **Submandibular gland** | Along the inner side of the mandible, in the floor of the mouth | Floor of the mouth, beside the lingual frenulum |
| **Sublingual gland** | Beneath the tongue, above the submandibular gland | Floor of the mouth, through several small ducts |

## The liver, gallbladder and pancreas

**Right and left lobes**
The two main lobes of the liver, separated in front by the falciform ligament

**Ligamentum teres (round ligament)**
A fibrous cord in the free edge of the falciform ligament; the remnant of the fetal umbilical vein

**Hepatic sinusoids**
Wide, leaky capillaries between the plates of hepatocytes

**Central vein**
The vein in the center of each liver lobule

**Head, body and tail of the pancreas**
The head sits in the curve of the duodenum and the tail reaches toward the spleen

**Accessory pancreatic duct**
Opens into the duodenum on its own

**Pancreatic islets**
Endocrine clusters that make hormones, about 1 percent of the pancreas

**Quadrate and caudate lobes**
Two small lobes on the inferior and posterior surface of the liver

**Hepatocytes**
The liver cells, arranged in plates

**Portal triad**
A bile duct, a branch of the hepatic artery and a branch of the hepatic portal vein at the corner of a lobule

**Gallbladder**
A pear-shaped sac on the inferior surface of the liver; fundus, body and neck

**Main pancreatic duct**
Joins the common bile duct at the hepatopancreatic ampulla

**Acini**
Exocrine clusters that make pancreatic juice, about 99 percent of the pancreas

## Veins of the abdomen and pelvis

Name the side, and say artery or vein, every time.

| VEIN | WHERE IT DRAINS |
|---|---|
| **External iliac vein** | Continues from the femoral vein and drains the lower limb; joins the internal iliac vein |
| **Internal iliac vein** | Drains the pelvic organs |
| **Right and left common iliac veins** | Formed by the internal and external iliac veins; they join to form the inferior vena cava |
| **Gonadal (testicular or ovarian) veins** | The right drains into the inferior vena cava; the left drains into the left renal vein |
| **Renal veins** | Drain the kidneys into the inferior vena cava |
| **Hepatic veins** | Drain the liver into the inferior vena cava |
| **Inferior vena cava** | Carries blood from the lower body to the right atrium |
| **Hepatic portal vein** | Does not drain into the inferior vena cava; it carries blood from the digestive organs into the liver |

## Fetal remnants in the abdomen

**Ligamentum teres (round ligament)**
Remnant of the umbilical vein, in the free edge of the falciform ligament

**Medial umbilical ligament**
Remnant of the umbilical artery, on the inside of the anterior abdominal wall

## Nerves and vessels of the lower limb

In the femoral triangle, from lateral to medial: femoral nerve, femoral artery, femoral vein.

| STRUCTURE | NERVE, ARTERY OR VEIN | WHERE IT RUNS | WHAT IT SERVES |
|---|---|---|---|
| **Femoral nerve** | Nerve, from the lumbar plexus (L2 to L4) | Under the inguinal ligament into the femoral triangle, lateral to the femoral artery | Anterior thigh muscles (quadriceps femoris, sartorius, iliacus, pectineus) and the skin of the front of the thigh; its saphenous branch supplies the skin of the medial leg |
| **Sciatic nerve** | Nerve, from the sacral plexus (L4 to S3); the largest nerve in the body | Leaves the pelvis through the greater sciatic foramen, deep to the gluteus maximus, then down the posterior thigh deep to the hamstrings; splits just above the knee | The hamstrings and part of the adductor magnus, and through its two branches all the muscles of the leg and foot and most of their skin |
| **Tibial nerve** | Nerve, a branch of the sciatic nerve | Straight down through the popliteal fossa and the posterior leg, then behind the medial malleolus into the sole | Posterior leg muscles (gastrocnemius, soleus, plantaris, tibialis posterior, flexor digitorum longus, flexor hallucis longus) and the sole of the foot |
| **Common fibular nerve** | Nerve, a branch of the sciatic nerve | Along the lateral edge of the popliteal fossa, then around the neck of the fibula just under the skin; splits into the superficial and deep fibular nerves | Lateral leg muscles (superficial fibular nerve), anterior leg muscles (deep fibular nerve), and the skin of the lateral leg and top of the foot |
| **Femoral artery** | Artery, the continuation of the external iliac artery past the inguinal ligament | Through the femoral triangle and down the front and medial thigh, then through the adductor hiatus to the back of the knee, where it becomes the popliteal artery | The thigh, mainly through its deep femoral branch, and the rest of the limb through the popliteal artery; the femoral pulse is felt in the groin |
| **Femoral vein** | Deep vein | Beside the femoral artery, medial to it in the femoral triangle; becomes the external iliac vein past the inguinal ligament | Drains the deep veins of the leg and thigh, and receives the great saphenous vein |
| **Great saphenous vein** | Superficial vein; the longest vein in the body | Begins on the medial side of the foot, passes in front of the medial malleolus, then up the medial leg and thigh to join the femoral vein near the groin | Drains the skin and superficial tissue of the medial leg and thigh; often taken for coronary bypass grafts |
| **Posterior tibial artery** | Artery, a branch of the popliteal artery | Down the posterior leg deep to the calf muscles, with the tibial nerve, then behind the medial malleolus into the sole | Posterior leg muscles and the sole of the foot; its fibular branch supplies the lateral leg. Its pulse is felt behind the medial malleolus |

## The endocrine glands

| GLAND | LOCATION | TISSUE, ZONES OR PARTS TO IDENTIFY |
|---|---|---|
| **Hypothalamus** | Floor and walls of the third ventricle, in the diencephalon of the brain | Connects to the pituitary gland by a stalk, the infundibulum |
| **Pituitary gland** | In the sella turcica of the sphenoid bone, hanging below the hypothalamus | Anterior lobe (glandular epithelium) and posterior lobe (nervous tissue) |
| **Pineal gland** | Posterior roof of the third ventricle, in the brain | A small cone-shaped gland; identify it by its position |
| **Thyroid gland** | Anterior neck, just below the larynx, on the front and sides of the trachea | Right lobe, left lobe and isthmus. On the slide: thyroid follicles, colloid, follicular cells, parafollicular (C) cells |
| **Parathyroid glands** | Posterior surface of the thyroid lobes; usually four | Small round glands; identify them by their position behind the thyroid |
| **Thymus** | Mediastinum, above the heart and behind the sternum | Large in children; shrinks after puberty and becomes mostly fat |

| GLAND | LOCATION | TISSUE, ZONES OR PARTS TO IDENTIFY |
|---|---|---|
| **Adrenal (suprarenal) glands** | On the superior pole of each kidney | Adrenal cortex, with three zones from the surface inward: zona glomerulosa, zona fasciculata (the widest), zona reticularis. Adrenal medulla in the center, made of modified nervous tissue |
| **Pancreas** | Behind the stomach; the head sits in the curve of the duodenum | Pancreatic islets (endocrine, pale clusters) scattered among the acini (exocrine) |
| **Ovaries** | Female pelvis, one on each side of the uterus | Endocrine glands as well as reproductive organs |
| **Testes** | In the scrotum | Endocrine glands as well as reproductive organs |

## Vessels of the anterior neck

The common carotid artery and internal jugular vein run together in the carotid sheath, with the vein lateral to the artery.

| VESSEL | ARTERY OR VEIN | WHERE IT RUNS | WHAT IT SUPPLIES OR DRAINS |
|---|---|---|---|
| **Common carotid artery** | Artery | Up the neck beside the trachea and larynx; the right comes from the brachiocephalic trunk and the left from the aortic arch. Splits into the internal and external carotid arteries at about the top of the thyroid cartilage | The head and neck, including most of the brain |
| **Vertebral artery** | Artery | Branches from the subclavian artery and climbs through the transverse foramina of the cervical vertebrae, then enters the skull through the foramen magnum | The back of the brain and the spinal cord |
| **Internal jugular vein** | Vein | From the jugular foramen down the neck, lateral to the common carotid artery; joins the subclavian vein to form the brachiocephalic vein | Drains the brain, face and neck |

PART C

# Questions you could be asked on the lab exam.

At each station a structure will be pinned or pointed to. These are the kinds of questions that go with it. Practice answering them out loud with a partner, with the structure in front of you.

### The Respiratory System

1. Identify the structure at the pointer.
2. Name the paranasal sinus at the pointer and the bone it is named for.
3. Name the region of the pharynx at the pointer. Does it carry air, food, or both?
4. Name the laryngeal cartilage at the pointer. Is it hyaline or elastic cartilage?
5. Which laryngeal cartilage is the only complete ring in the airway?
6. Is the fold at the pointer a true vocal fold or a vestibular (false) fold? Which one makes sound?
7. Is the bronchus at the pointer the right or the left main bronchus? Why does an inhaled object most often end up on the right?
8. Name the lobe at the pointer and say whether this is the right or the left lung.
9. Name the fissure at the pointer: horizontal or oblique.
10. Is the membrane at the pointer the parietal pleura or the visceral pleura?
11. Name the vessel at the pointer, give its side if it has one, and say whether it is an artery or a vein.
12. Which nerve supplies the diaphragm, and what passes through the openings at T8, T10, and T12?

### The Lymphatic System

1. Identify the lymphatic structure at the pointer.
2. Which duct drains the right arm and the right side of the head, neck, and thorax, and which drains the rest of the body?
3. Name the sac at the pointer where the thoracic duct begins.
4. Where do the lymphatic ducts empty into the blood?
5. Name the part of the lymph node at the pointer: capsule, cortex, medulla, or hilum.
6. Are the vessels at the pointer afferent or efferent? Is lymph flowing into or out of the node?
7. Is the organ at the pointer a primary or a secondary lymphatic organ?
8. Name the tonsil at the pointer and the region of the pharynx it guards.
9. What are lacteals, where are they found, and what do they absorb?
10. Trace a drop of interstitial fluid from a tissue space back to the bloodstream.

### The Alimentary Canal

1. Identify the structure or region at the pointer.
2. Name the fold of peritoneum at the pointer and what it connects.
3. Is the organ at the pointer intraperitoneal or retroperitoneal? Name the retroperitoneal organs using SADPUCKER.
4. Name the layer of the wall of the canal at the pointer.
5. Which part of the esophagus has an adventitia instead of a serosa?
6. Name the region of the stomach at the pointer.
7. Which extra muscle layer does the stomach have, and where does it sit?
8. Name the part of the small intestine at the pointer: duodenum, jejunum, or ileum.
9. Identify the feature at the pointer: circular fold, villus, or microvilli. What do all three do?
10. Name the part of the colon or the flexure at the pointer.
11. Name the feature of the large intestine at the pointer: teniae coli, haustra, or omental appendices.
12. Trace a bite of food from the mouth to the anus, naming every region it passes through.

### The Accessory Digestive Organs

1. Identify the structure at the pointer.
2. Name the salivary gland at the pointer and say where its duct opens.
3. Name the lobe of the liver at the pointer.
4. Name the ligament at the pointer. The ligamentum teres is the remnant of which fetal structure?
5. Name the three parts of the portal triad. Which two bring blood in, and which one carries bile out?
6. On the liver slide, identify the structure at the pointer: hepatocytes, sinusoid, central vein, or portal triad.
7. Name the region of the gallbladder at the pointer: fundus, body, or neck.
8. Name the duct at the pointer.
9. Which two ducts join to form the common bile duct?
10. Name the region of the pancreas at the pointer. Which pancreatic duct joins the common bile duct?
11. On the pancreas slide, which clusters are exocrine and which are endocrine?
12. Trace a drop of bile from the hepatocyte that made it to the duodenum.

### Vessels of the Abdomen and Pelvis

1. Identify the vessel at the pointer, give its side if it has one, and say whether it is an artery or a vein.
2. Name the three unpaired branches of the abdominal aorta in order from superior to inferior.
3. What does the artery at the pointer supply?
4. Name the three branches of the celiac trunk.
5. Is the paired artery at the pointer a renal artery or a gonadal artery?
6. The abdominal aorta ends by dividing into which two arteries?
7. Is the artery at the pointer the internal or the external iliac artery? What does it supply?
8. The medial umbilical ligament is the remnant of which fetal vessel?
9. Which two veins join to form the hepatic portal vein, and which vein does the inferior mesenteric vein usually drain into?
10. Where do the hepatic veins empty?
11. What makes the hepatic portal system a portal system? Name the two capillary beds the blood passes through.
12. Trace a molecule of glucose absorbed by the small intestine from the intestinal capillary to the right atrium.

### Muscles, Nerves and Vessels of the Lower Limb

1. Identify the muscle at the pointer.
2. Give the origin, insertion and action of the muscle at the pointer.
3. Name the four muscles of the quadriceps femoris, and give their shared insertion and action.
4. Name the three hamstrings and the bone marking they arise from.
5. Is the muscle at the pointer in the anterior, medial, or posterior compartment of the thigh?
6. Which gluteal muscle is the main extensor of the hip, and which one abducts the thigh?
7. Identify the structure at the pointer: tensor fasciae latae or iliotibial band. Which two muscles pull on the band?
8. Is the leg muscle at the pointer in the anterior, lateral, or posterior compartment, and which nerve supplies it?
9. Which two muscles join to form the calcaneal (Achilles) tendon, and which one also flexes the knee?
10. Name the nerve at the pointer and the plexus or nerve it comes from.
11. Name the vessel at the pointer, give its side, and say whether it is an artery or a vein. If it is a vein, is it deep or superficial?
12. Name the three structures in the femoral triangle in order from lateral to medial.

### The Endocrine System

1. Identify the gland at the pointer and give its location.
2. Name the part of the thyroid gland at the pointer: right lobe, left lobe, or isthmus.
3. Where are the parathyroid glands found?
4. On the thyroid slide, identify the structure at the pointer: thyroid follicle, colloid, follicular cell, or parafollicular (C) cell.
5. Is the region at the pointer the adrenal cortex or the adrenal medulla?
6. Name the zone of the adrenal cortex at the pointer, and give all three zones in order from the surface inward.
7. Which gland sits on the superior pole of each kidney?
8. On the pancreas slide, are the cells at the pointer a pancreatic islet or acini? Which one is endocrine?
9. Which gland sits in the sella turcica of the sphenoid bone, and which part of the brain is it attached to?
10. Which endocrine gland lies in the mediastinum and shrinks after puberty?

### The Anterior Neck

1. Identify the structure at the pointer.
2. Give the origin, insertion, action and nerve of the sternocleidomastoid.
3. What happens to the head when only the right sternocleidomastoid contracts?
4. Is the vessel at the pointer an artery or a vein? Give its side.
5. Which artery climbs the neck through the transverse foramina of the cervical vertebrae?
6. Which vein runs beside the common carotid artery, and which of the two lies more lateral?
7. Which vein does the internal jugular vein join to form the brachiocephalic vein?